

หลักสูตรอบรม

# Basic to Intermediate React.js



React 19 + TypeScript + Vite

พัฒนาเว็บแอปพลิเคชันยุคใหม่ด้วยเครื่องมือมาตรฐานปี 2026  
ครบตั้งแต่พื้นฐานถึงการนำขึ้นใช้งานจริง (Deployment)



React 19



TypeScript



Vite 6



Tailwind  
CSS v4



React  
Router v7



TanStack  
Query



Zustand



Redux  
Toolkit



React  
Hook Form



Zod

วิทยากร

**อาจารย์สามิตร ไทยม**

IT Genius Engineering Co., Ltd.



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# Basic to Intermediate React.js

วิทยากร



**อ.สามิตร โกยม (ปาน)**

ปริญญาโท คณะเทคโนโลยีและสารสนเทศ  
สถาบันเทคโนโลยีพระจอมเกล้าพระนครเหนือ

## ▶ Frontend

Angular, React, Vue, Next, Nuxt, Bootstrap, Tailwind CSS

## ▶ Backend

PHP, Python, Java, Kotlin, Go, Rust, NodeJS, NestJS, .NET

## ▶ Database

MySQL, PostgreSQL, MS SQL Server, MongoDB, Supabase

## ▶ Mobile

Java, Kotlin, Objective C, Swift, KMP, Cordova, Flutter, React Native, Expo

## ▶ DevOps

Git, Github, Gitlab, Docker, Kubernetes, Jenkins, CI/CD

# แบบทดสอบก่อนอบรม

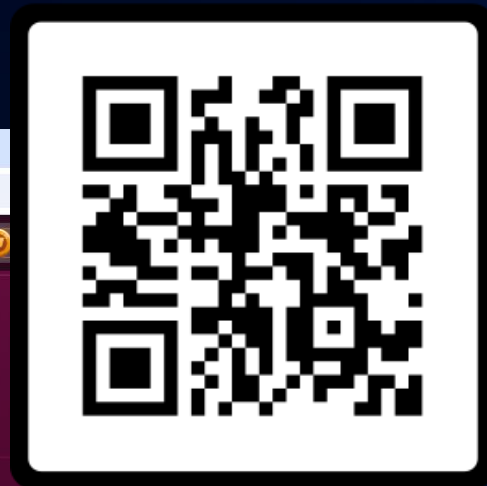
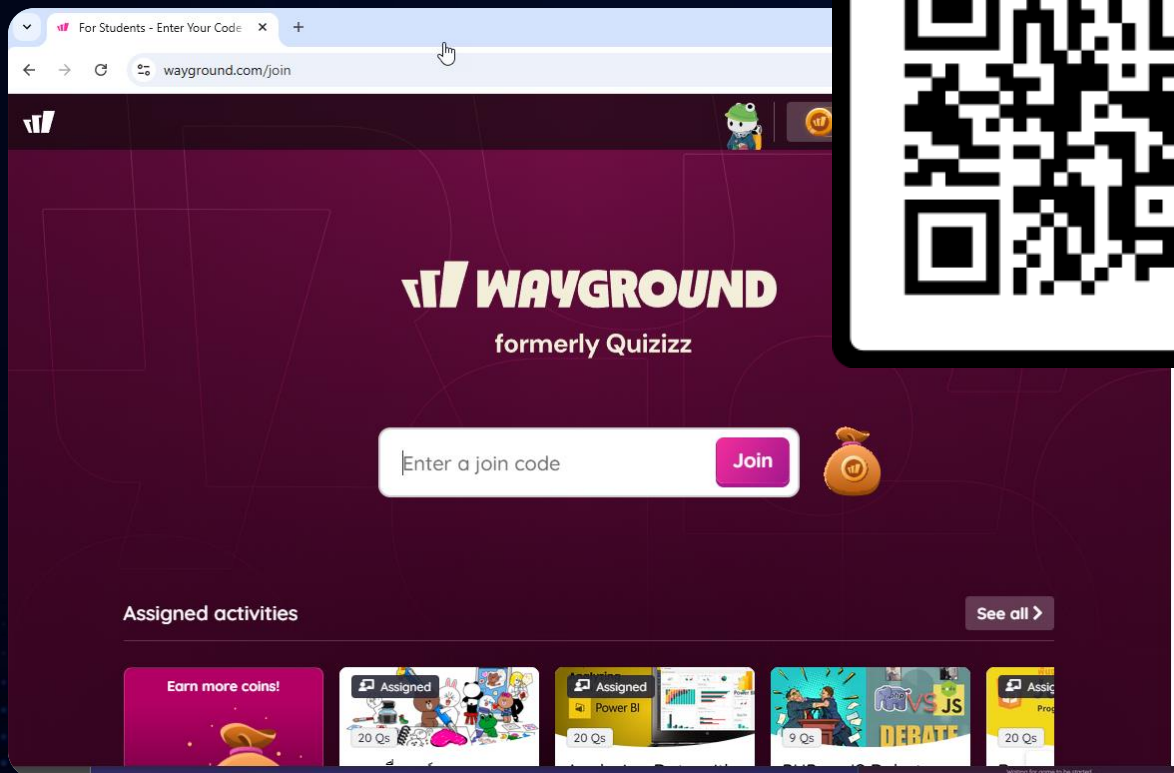




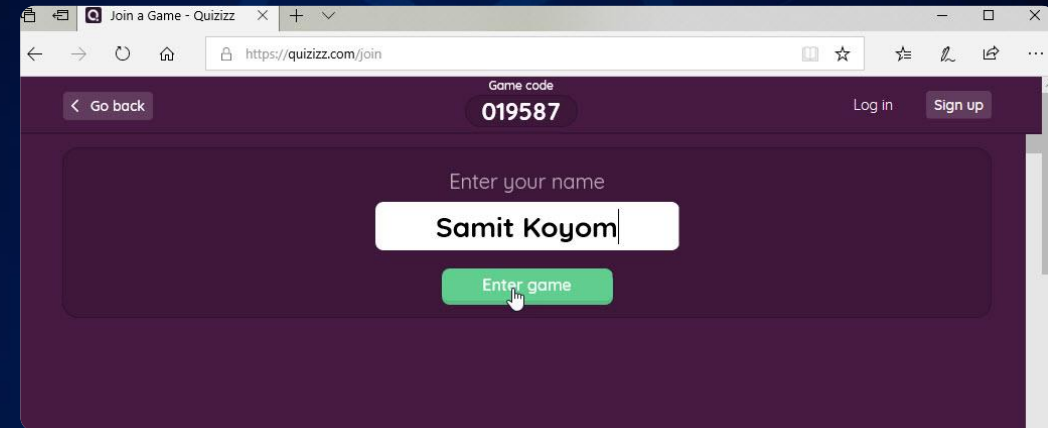
# Pretest ทำแบบทดสอบก่อนอบรม

**STEP 1:** เข้าทำแบบทดสอบที่ลิงค์ ป้อนรหัสเข้าห้องสอบ

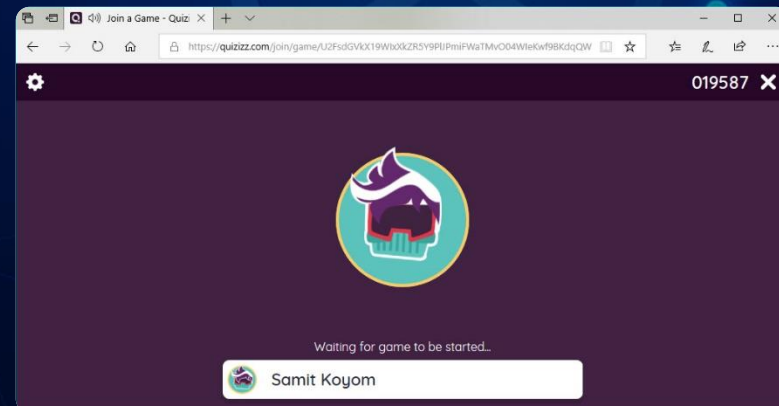
[wayground.com/join](https://wayground.com/join)



**STEP 2:** ป้อนชื่อ



**STEP 3:** รอผู้สอน Start ข้อสอบ



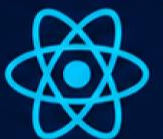
# Basic to Intermediate React.js

ดาวน์โหลดเอกสารประกอบการอบรม

**<https://bit.ly/itreact>**

หลักสูตรอบรม

# Basic to Intermediate React.js



React 19 + TypeScript + Vite

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# Day 1 — Agenda

0

## Setup

Node.js, VS Code Extensions, Chrome DevTools

1.1

## React Intro

Virtual DOM, Reconciliation, Ecosystem

1.2

## Vite + Tools

CRA vs Vite, HMR, ESLint, Prettier

1.3

## First Project

Vite + React + TypeScript, File Structure

1.4

## TypeScript

Types, Interface, Generics, Union Types

1.5

## JSX & Component

JSX Rules, Function Component, Fragment

[WS]

## Workshop

Build "Employee Card" App

# โปรแกรมที่ใช้อบรม

1. Visual Studio Code
2. Node.JS 22.x Up
3. Git



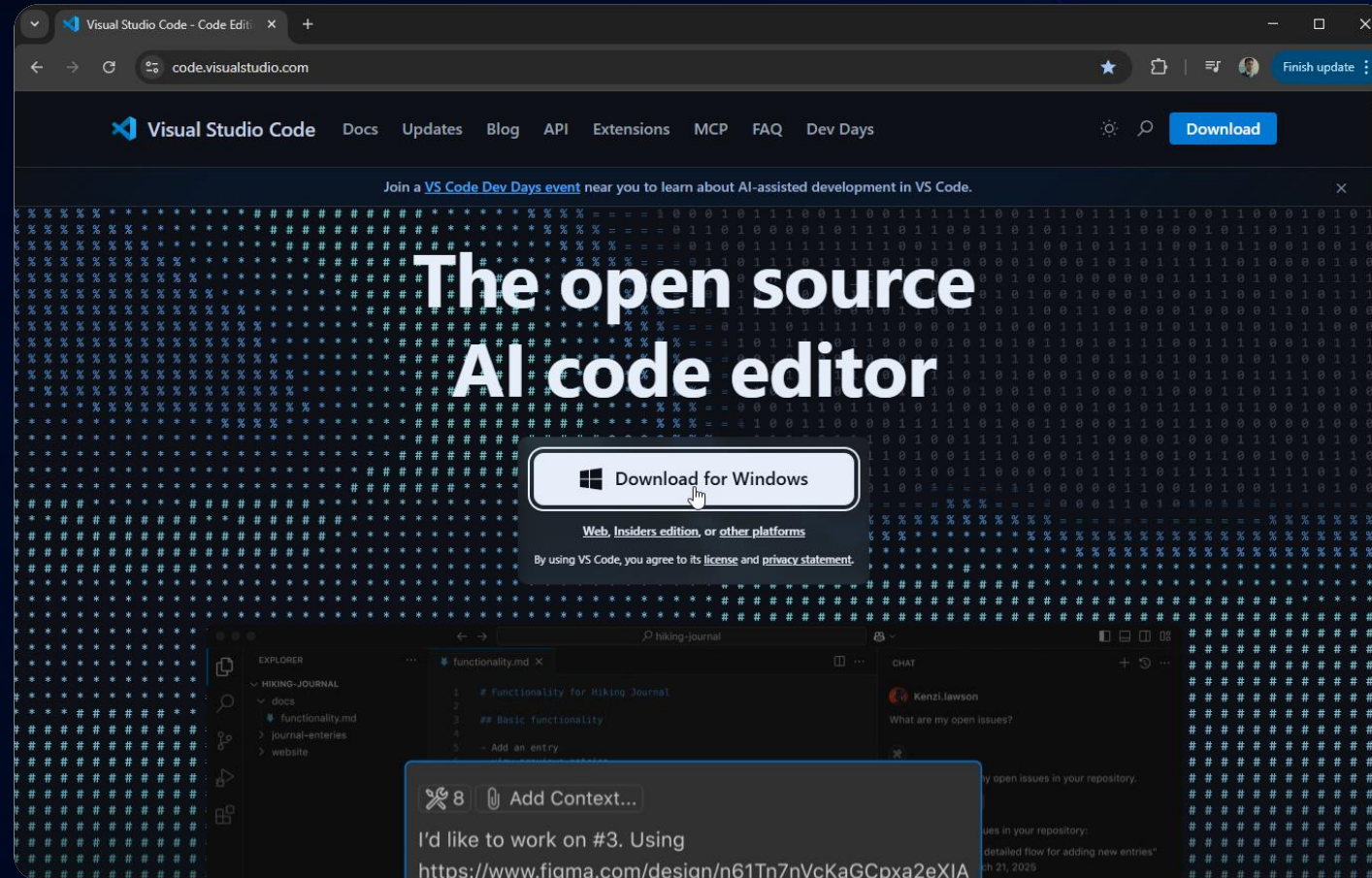




# 1. ติดตั้ง Visual Studio Code



# ติดตั้ง Visual Studio Code พร้อมส่วนเสริมที่จำเป็น



เข้าไปดาวน์โหลด Visual Studio Code ได้ที่ <https://code.visualstudio.com>

# การติดตั้งส่วนเสริม (Extension) ของ Visual Studio Code



File Edit Selection View Go Run Terminal Help

EXTENSIONS: MARKETPLACE

Auto Import

**Auto Import - ES6, TS, ...** 814K ★ 3.5  
Automatically finds, parses and provide...  
Sergey Korenuk **Install**

**Auto Import**  
Automatically finds, parses and provide...  
steoates **Install**

**Auto Import - ES6 & TS** 131K ★ 3.5  
Automatically finds, parses and provide...  
Martin Oppitz **Install**

**Flutter-Auto-Import** 108K ★ 3  
auto import flutter's package  
davidwoo **Install**

**Svelte Auto Import** 69K ★ 5  
Automatically finds, parses and provide...  
pivaszbs **Install**

**Auto Import Relative Path** 54K ★ 5  
Streamline your imports with instant rel...

Extension: Auto Import - ES6, TS, JSX, TSX



## Auto Import - ES6, TS, JSX, TSX

Sergey Korenuk

814,468

★★★★☆ (29)

Automatically finds, parses and provides code actions and code completion for all

**Install**



Auto Update



DETAILS

FEATURES

## Auto Import - ES6, TS, JSX, TSX (VSCode Extension)

Automatically finds, parses and provides code actions and code completion for all available imports. Works with Java and TypeScript (TS). Forker from old repo [vscode-extension-auto-import](#)

```
widget-group.component.ts
1 import { WidgetGroup } from '../shared/widget-group';
2 import { WidgetComponent } from '../shared/widget/widget.component';
3 import { Component, Input } from '@angular/core';
4
5 @Component({
6   selector: 'exp-dashboard-widget-group',
7   directives: [WidgetComponent],
8   templateUrl: './exp/modules/dashboard/widget-list-view/widget-group/widget-group.html',
9   styleUrls: ['./exp/modules/dashboard/widget-list-view/widget-group/widget-group.css']
10 })
11
12 export class WidgetGroupComponent {
```

# รายชื่อ Extensions ที่แนะนำสำหรับ VS Code

1. **Auto Import – ES6, TS, JSX, TSX** by Sergey Korenuk
2. **ES7 + React/Redux/React-Native snippet** by dsznajder
3. **Markdown Preview Enhanced** by Yiyi Wang
4. **MySQL** by Database Client
5. **Tailwind CSS IntelliSense** by Tailwind Labs
6. **Catppuccin for VSCode** by Catppuccin
7. **Catppuccin Icons for VSCode** by Catppuccin



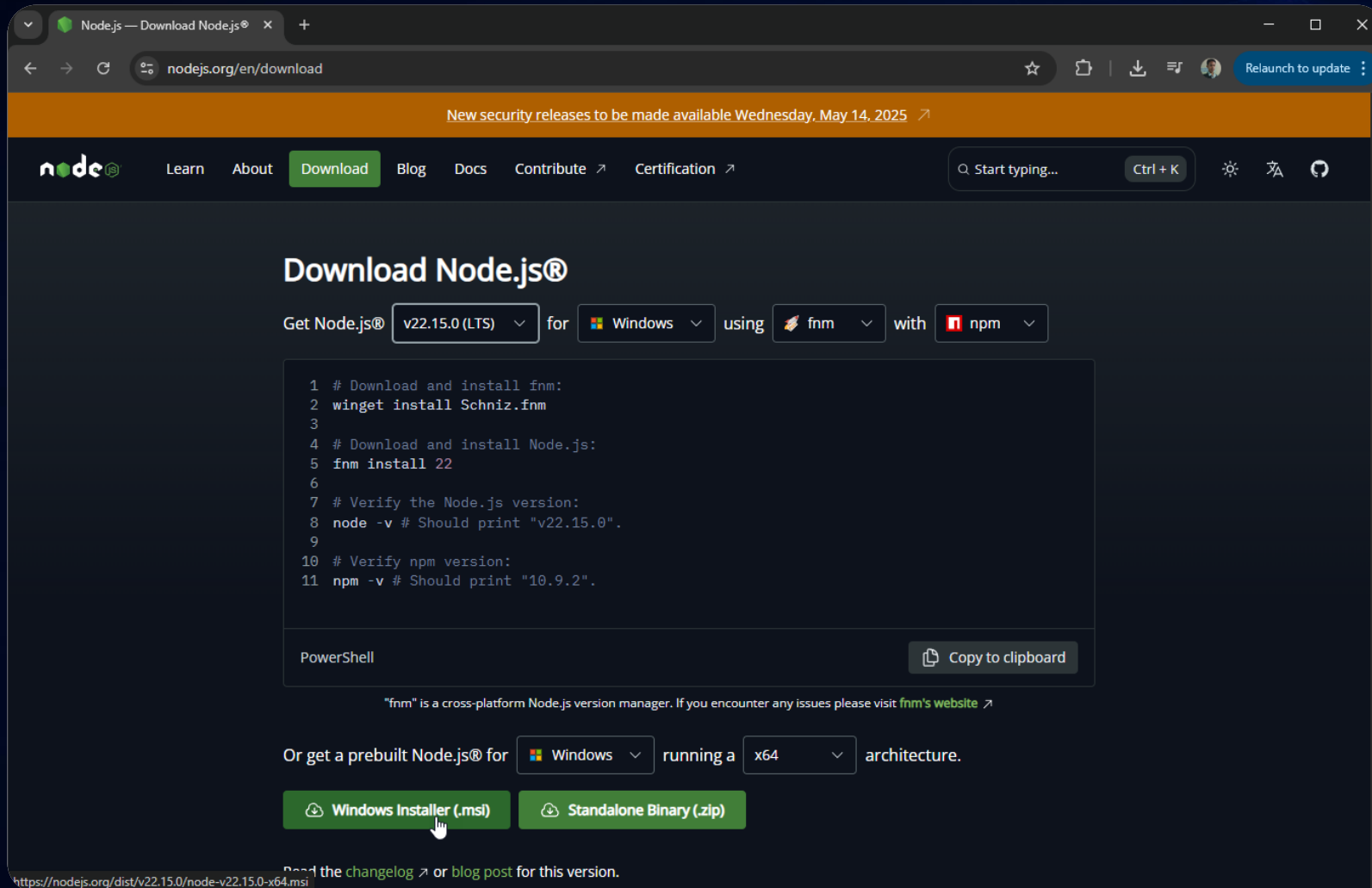
## 2. ติดตั้ง Node JS





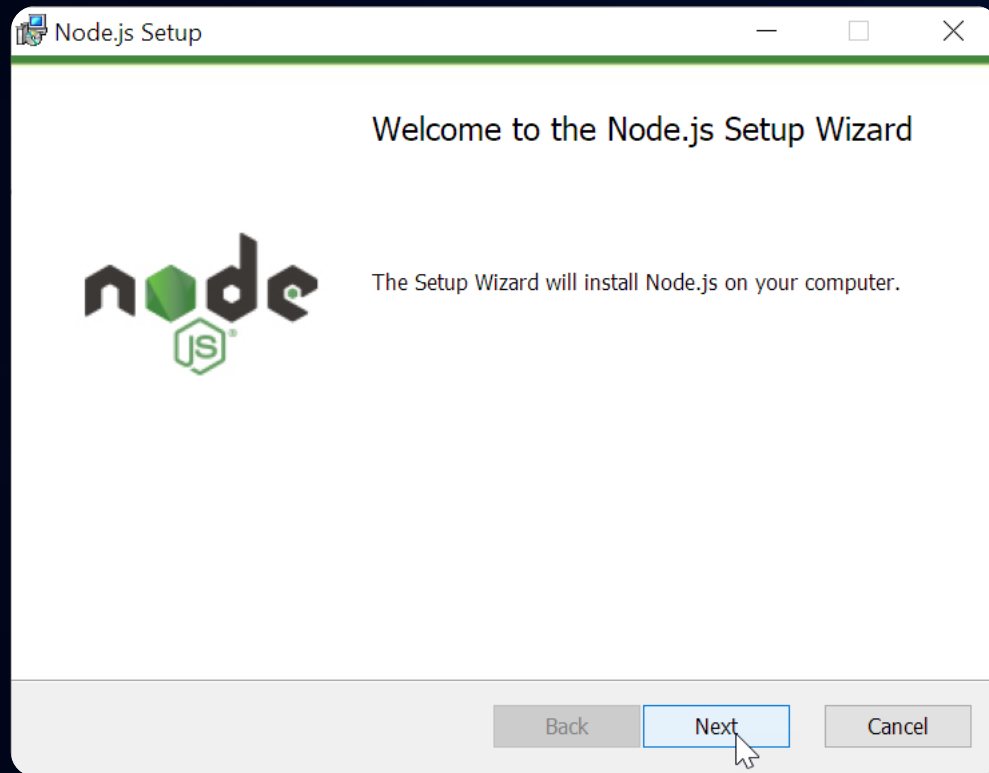
# Download Node.JS V.22.x

<https://nodejs.org/en/>

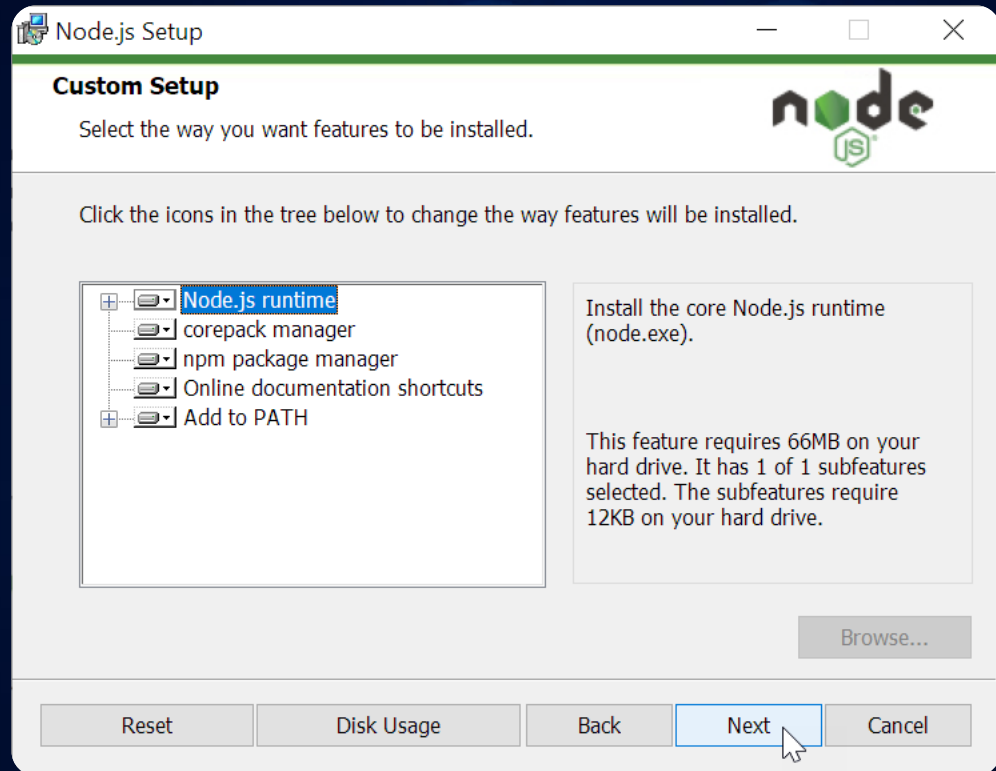
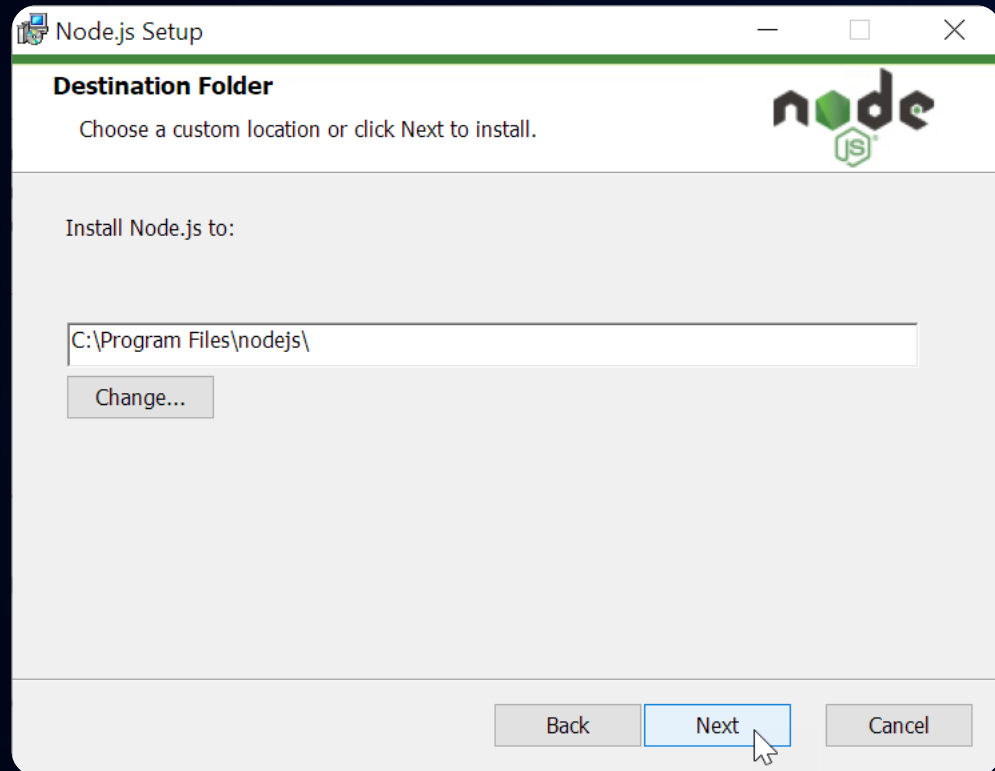


หมายเหตุ การอบรมรองรับตั้งแต่ Node.JS 20 ขึ้นไป สามารถใช้ Node.JS 21 , 22, 23 หรือ 24 ก็ได้

## ติดตั้ง Node.JS V.22.x

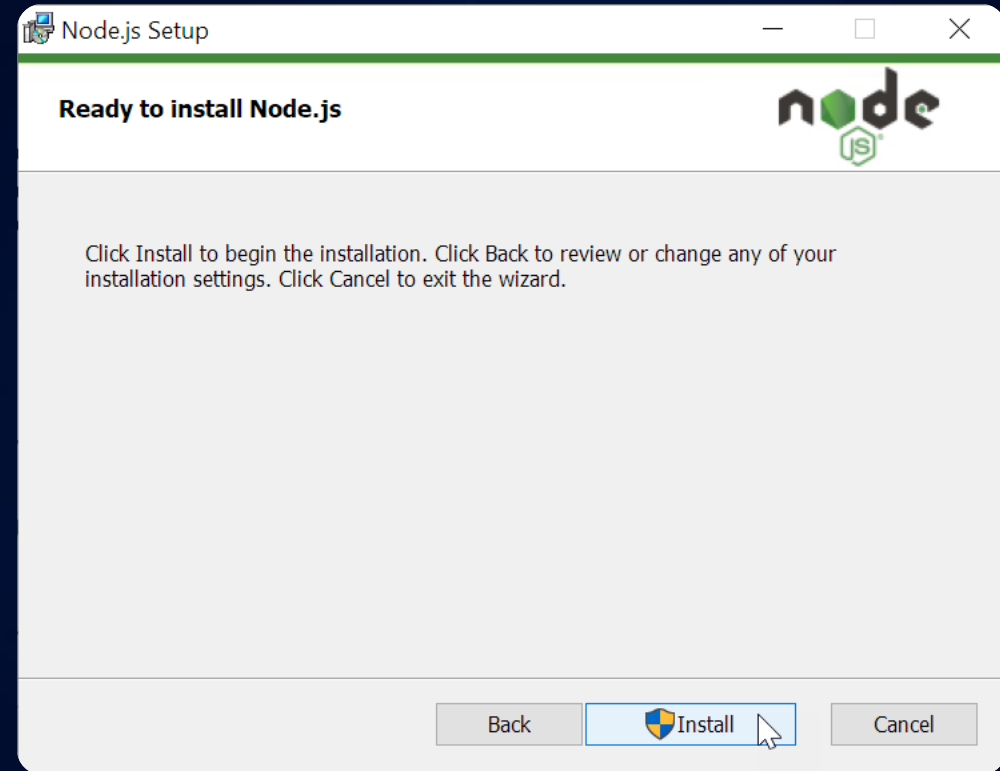
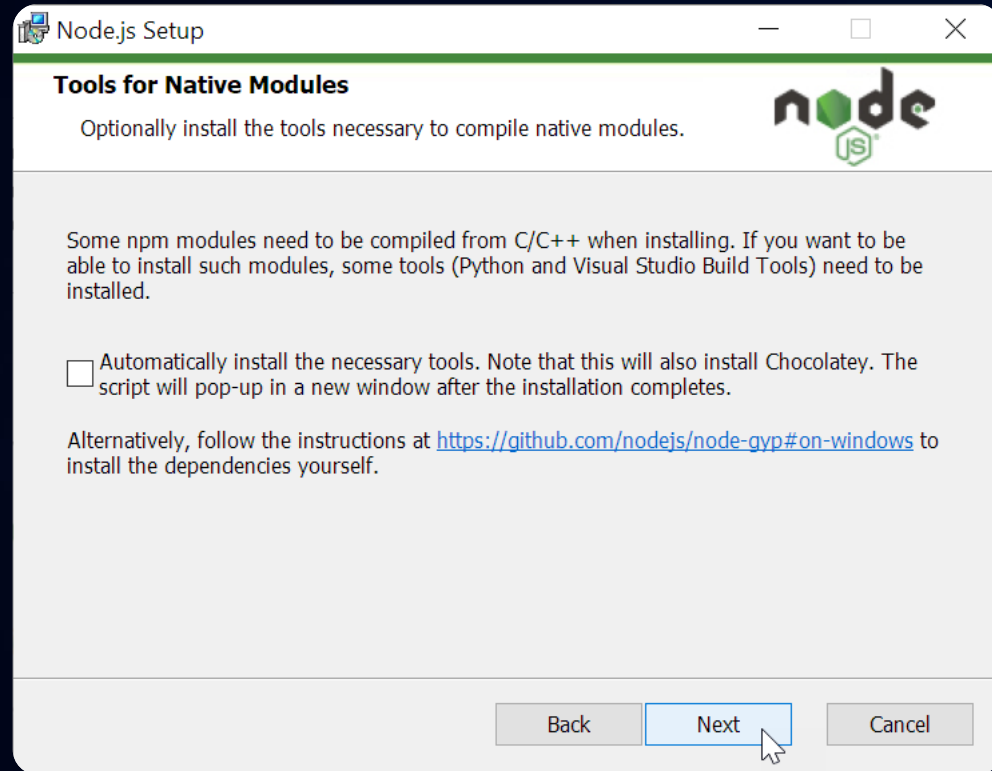


## ติดตั้ง Node.JS V.22.x

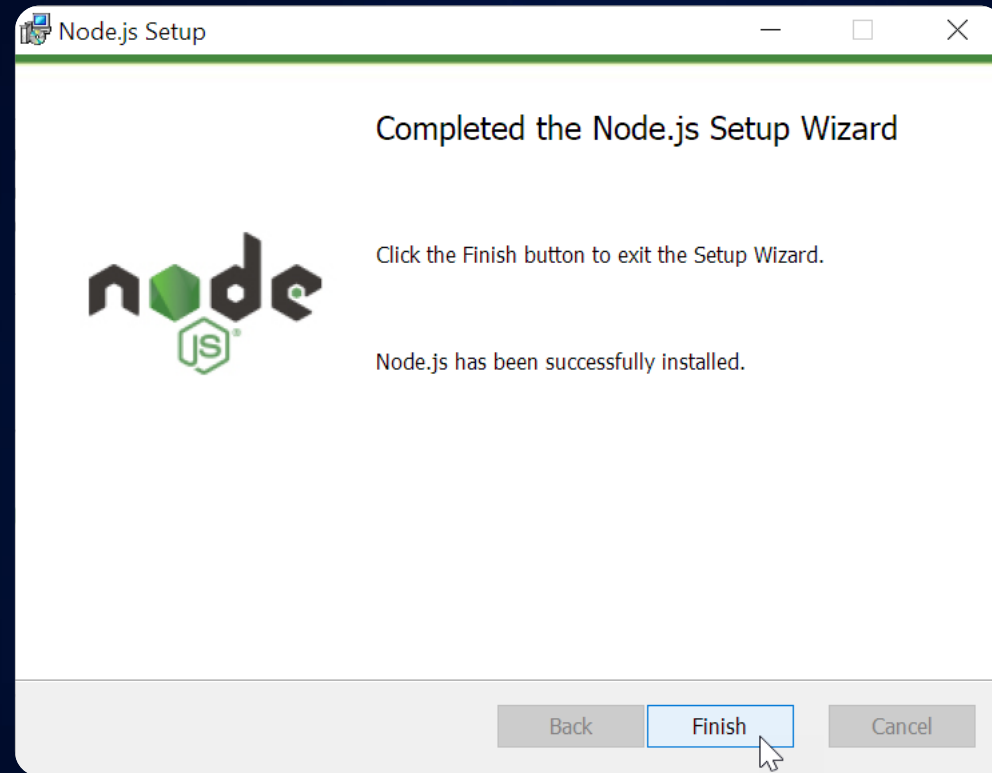
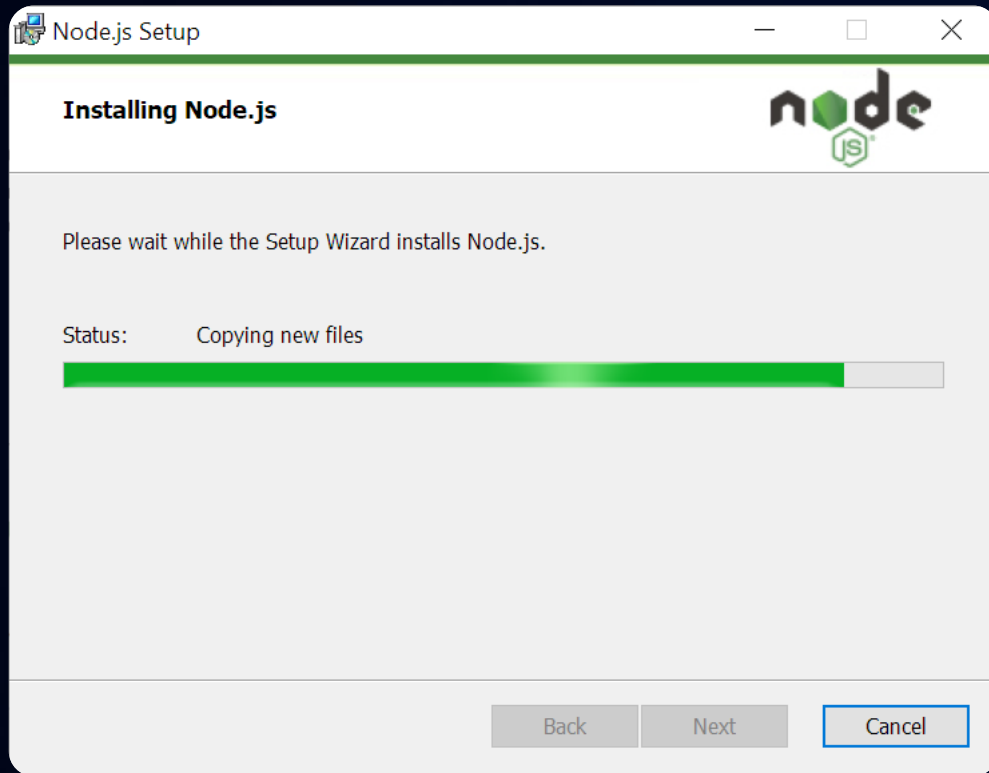




## ติดตั้ง Node.JS V.22.x



## ติดตั้ง Node.JS V.22.x



ทดสอบหลังติดตั้งเสร็จ

node -v

```
C:\Users\Samit>node -v  
v22.14.0
```

```
C:\Users\Samit>|
```

npx -v

```
C:\Users\Samit>npx -v  
10.9.2
```

```
C:\Users\Samit>|
```

npm -v

```
C:\Users\Samit>npm -v  
10.9.2
```

```
C:\Users\Samit>|
```

**หมายเหตุ** การยอมรับรองรับตั้งแต่ Node.JS 20 ขึ้นไป สามารถใช้ Node.JS 21 , 22, 23 หรือ 24 ก็ได้

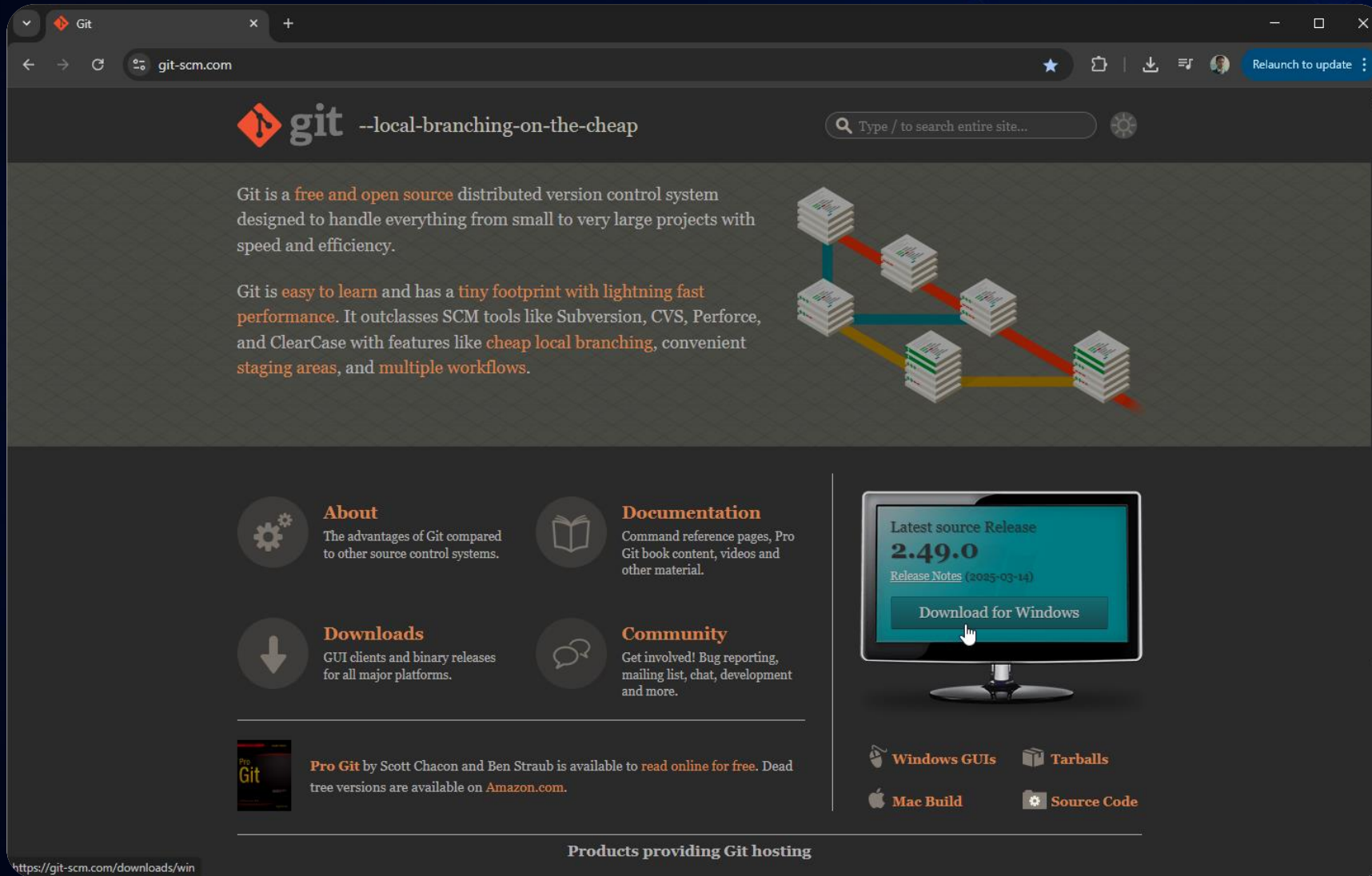




### 3. ติดตั้ง Git



ดาวน์โหลดไฟล์ติดตั้ง Git ได้ที่ <https://git-scm.com/>



The screenshot shows the Git website homepage in a web browser. The browser's address bar displays 'git-scm.com'. The page features the Git logo and the tagline '--local-branching-on-the-cheap'. A search bar is located in the top right. The main content area describes Git as a free and open source distributed version control system, highlighting its speed, efficiency, ease of learning, and performance. A diagram of a branching model is shown to the right. Below this, there are four sections: 'About', 'Documentation', 'Downloads', and 'Community'. The 'Downloads' section includes a 'Latest source Release' of 2.49.0 and a 'Download for Windows' button. At the bottom, there are links for 'Windows GUIs', 'Tarballs', 'Mac Build', and 'Source Code'. The footer mentions 'Products providing Git hosting'.

Git --local-branching-on-the-cheap

Git is a **free and open source** distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is **easy to learn** and has a **tiny footprint with lightning fast performance**. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like **cheap local branching**, convenient staging areas, and **multiple workflows**.

**About**  
The advantages of Git compared to other source control systems.

**Documentation**  
Command reference pages, Pro Git book content, videos and other material.

**Downloads**  
GUI clients and binary releases for all major platforms.

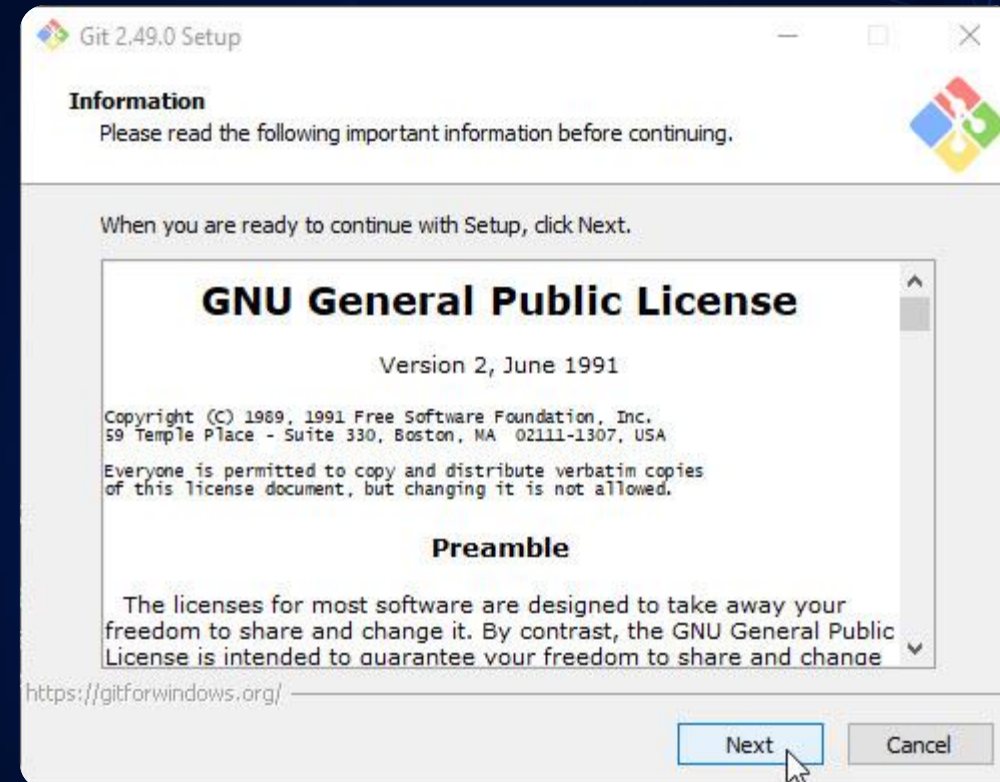
**Community**  
Get involved! Bug reporting, mailing list, chat, development and more.

**Latest source Release**  
**2.49.0**  
[Release Notes \(2025-03-14\)](#)  
[Download for Windows](#)

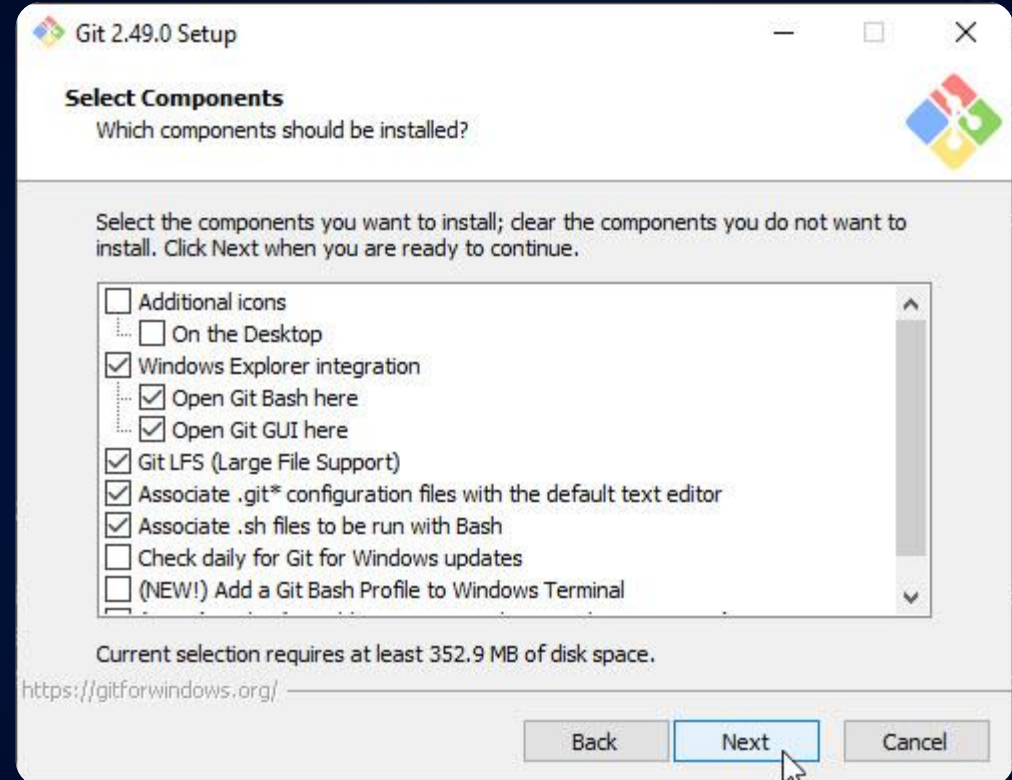
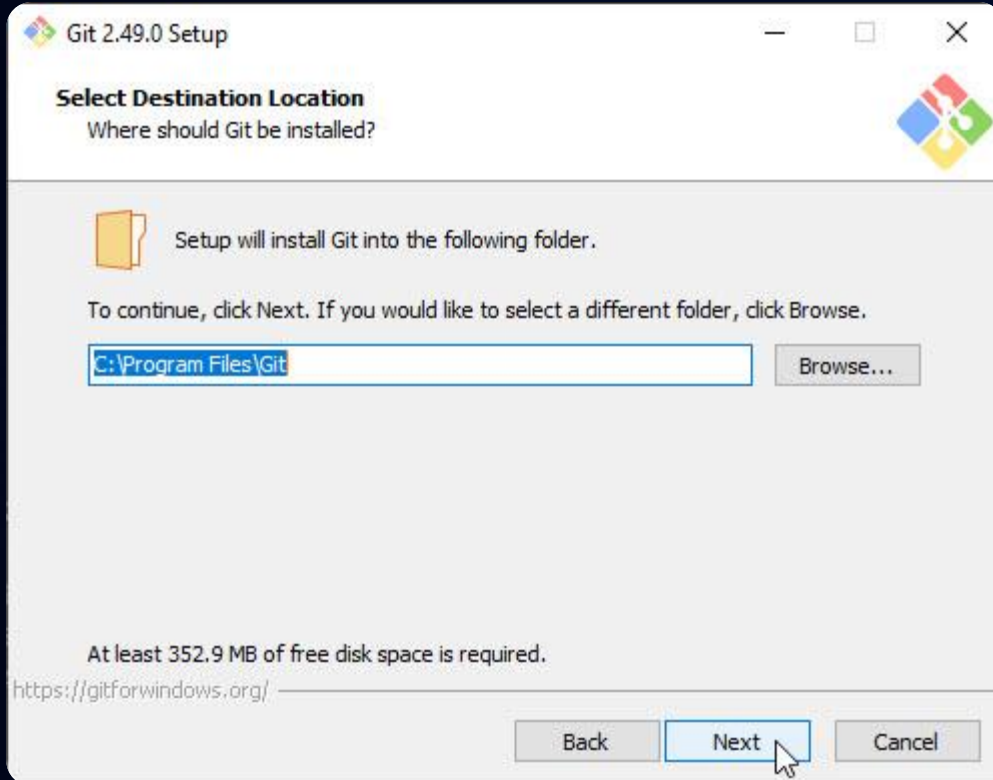
**Windows GUIs** **Tarballs**  
**Mac Build** **Source Code**

Products providing Git hosting

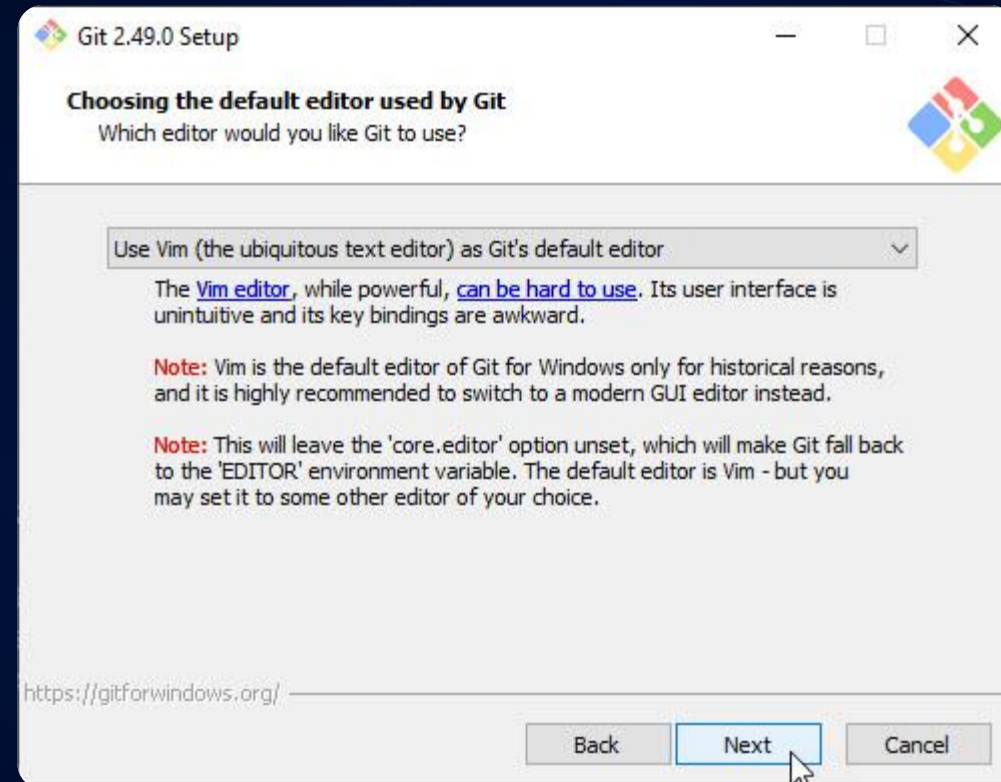
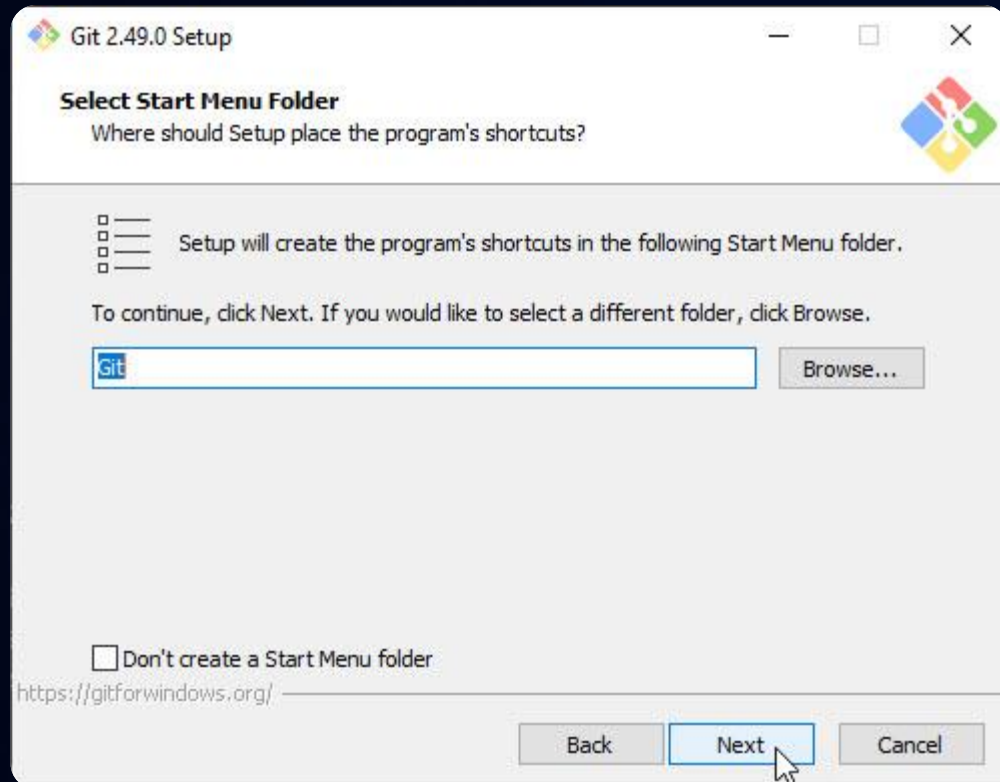
<https://git-scm.com/downloads/win>

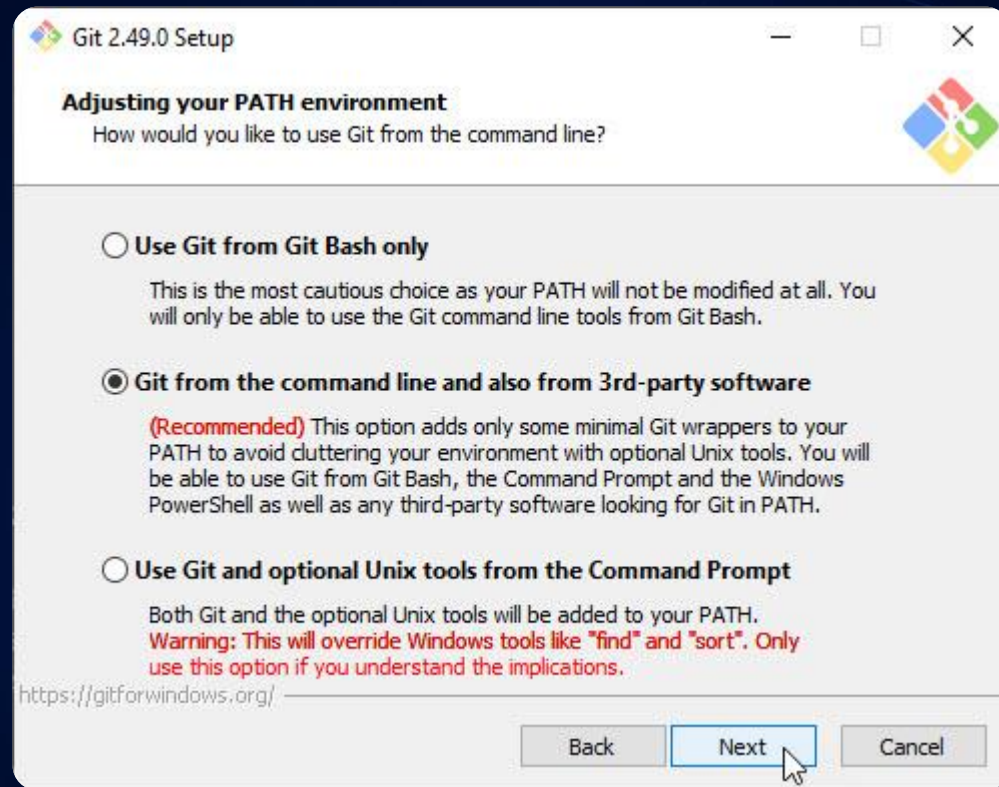
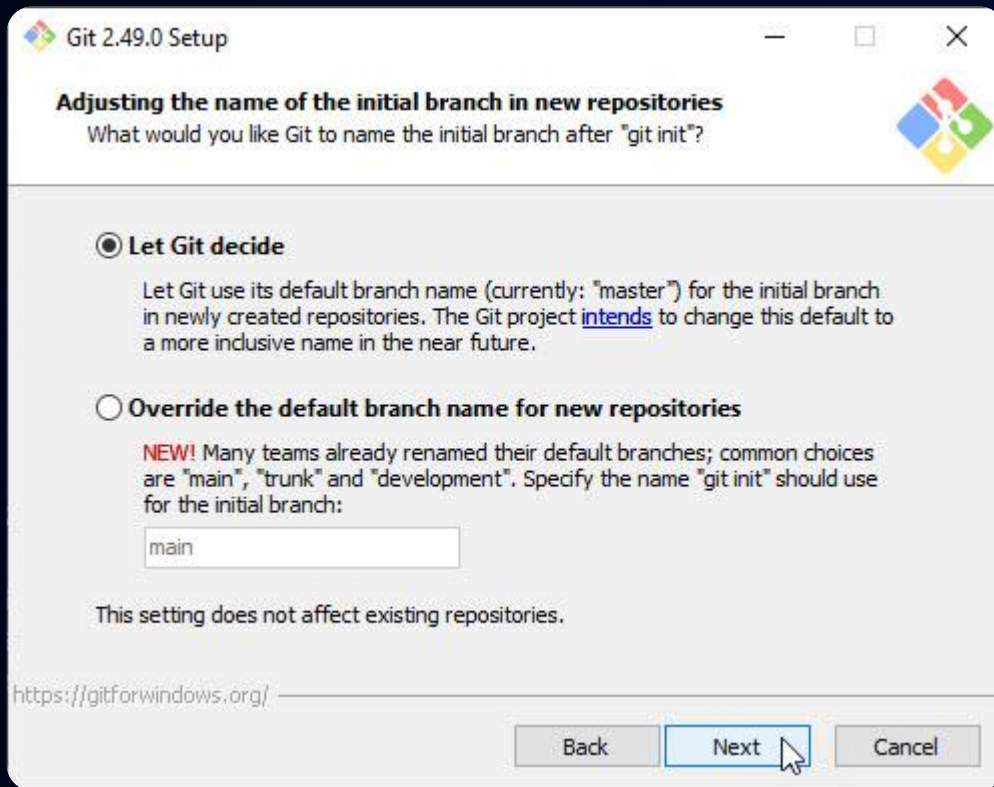


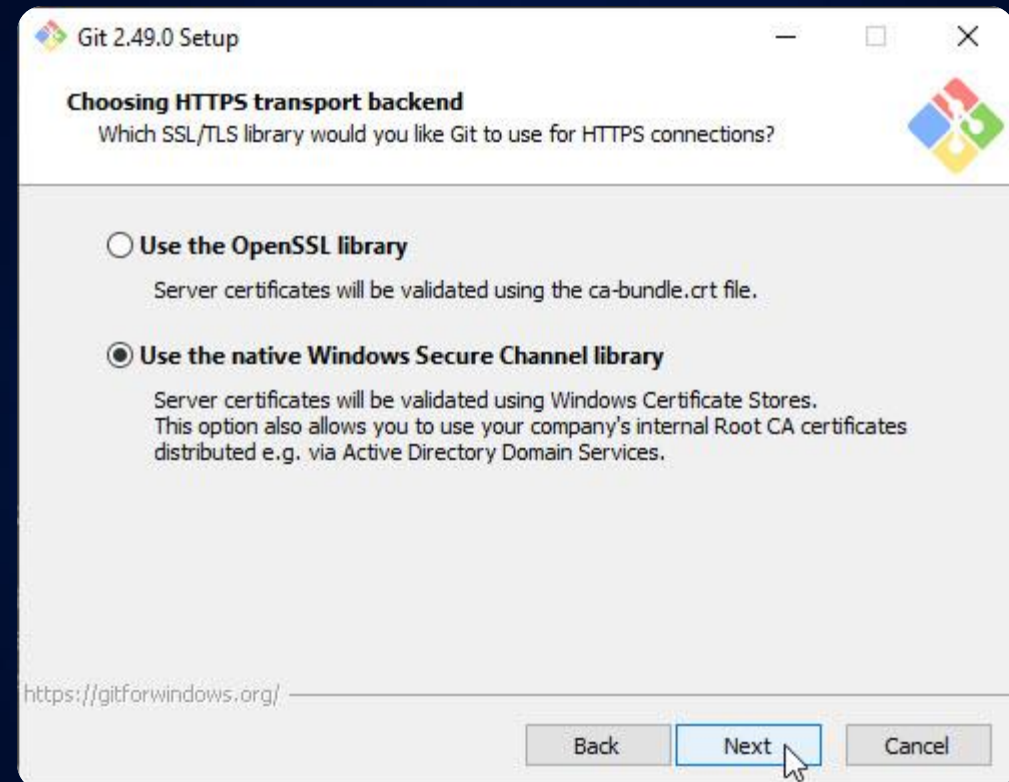
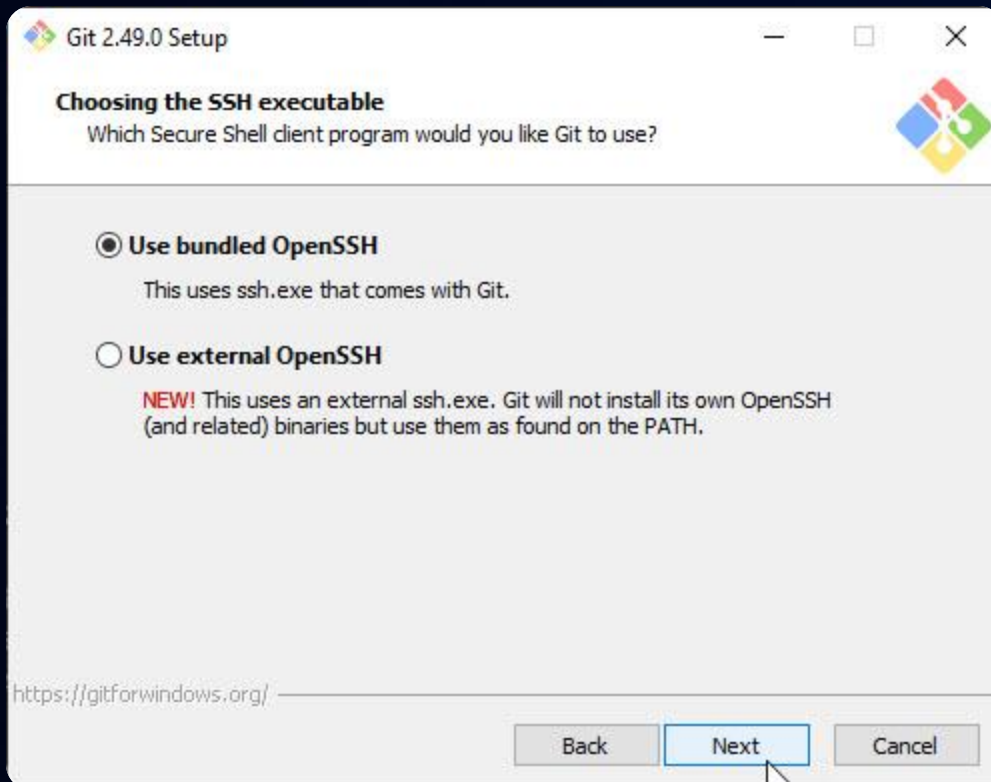




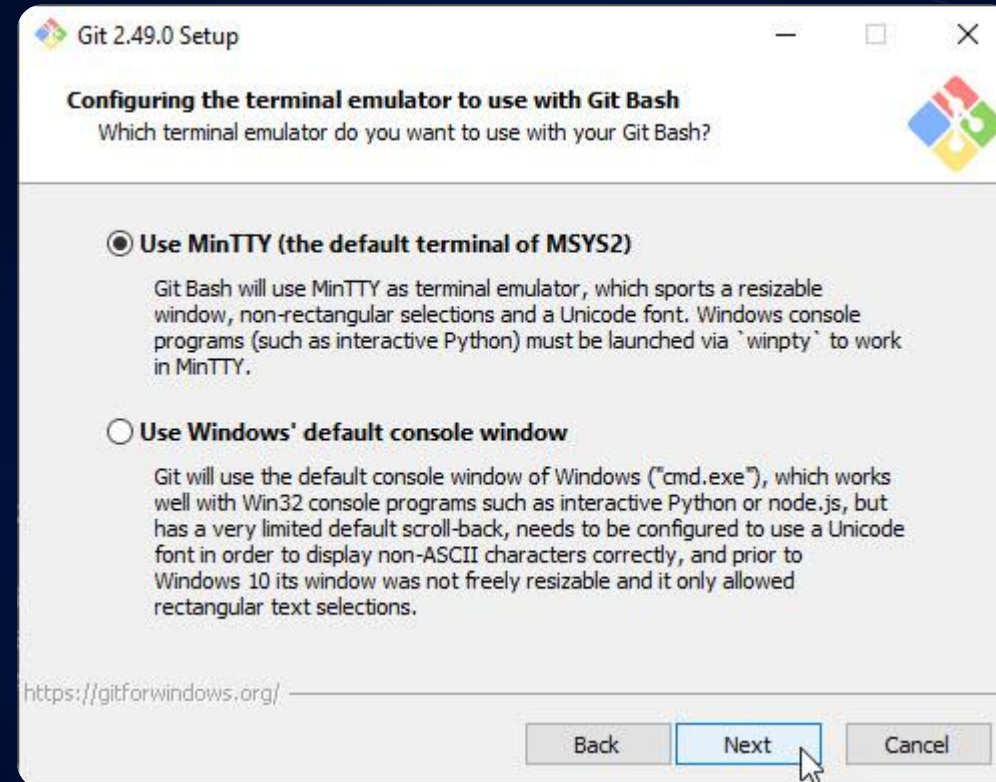
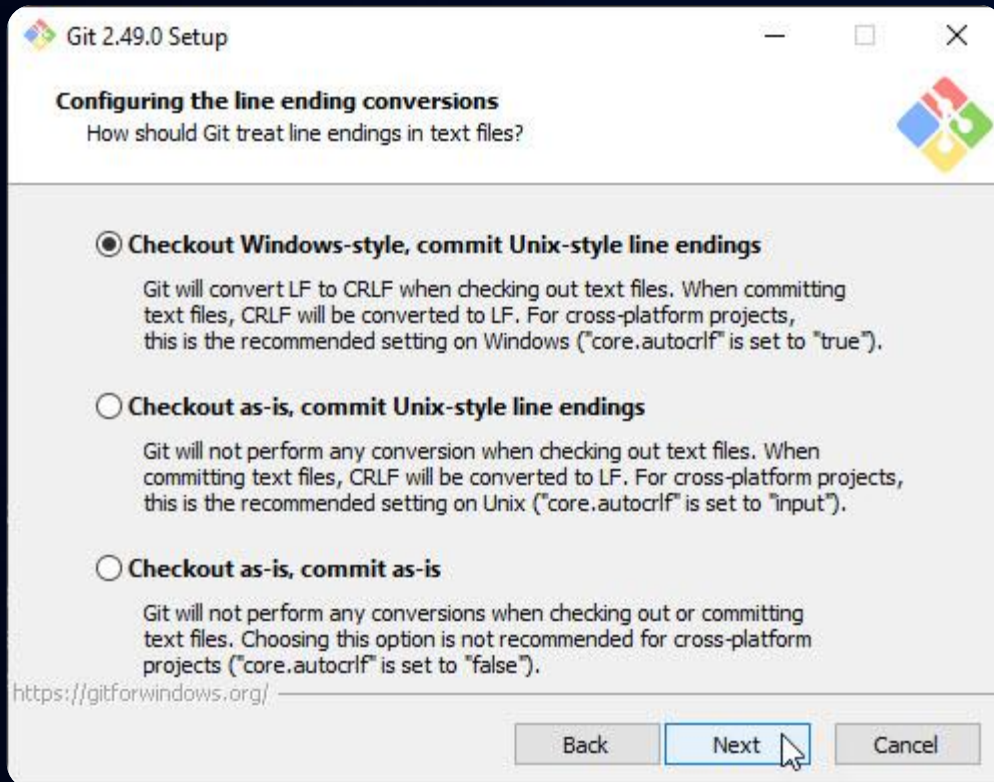


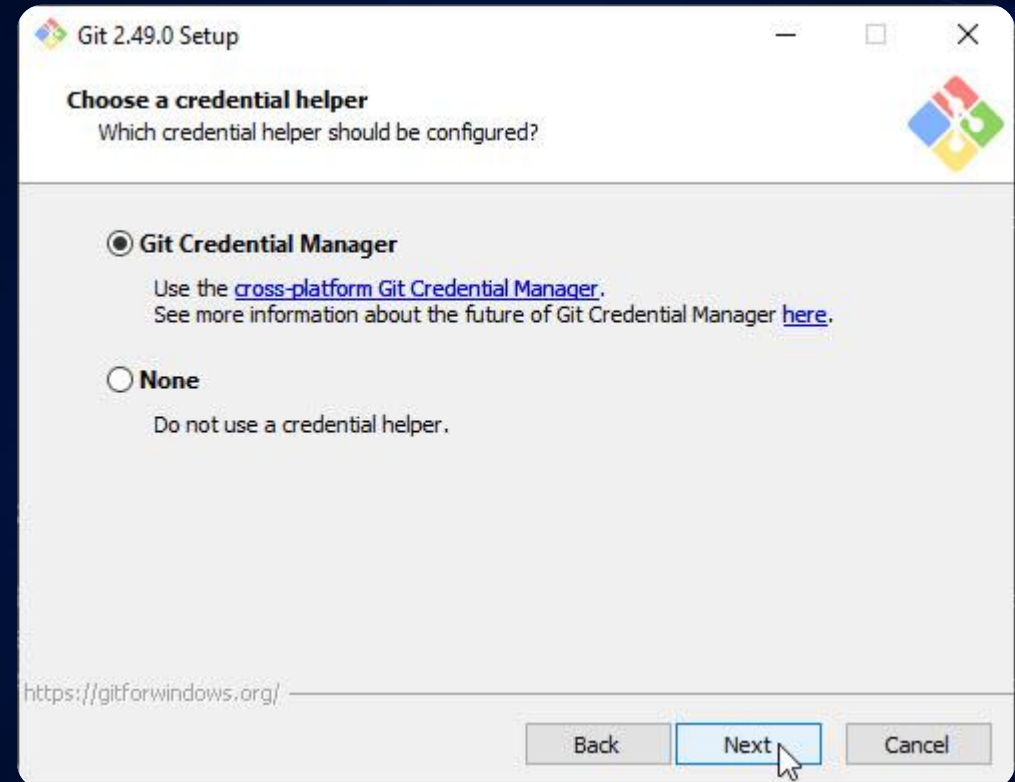
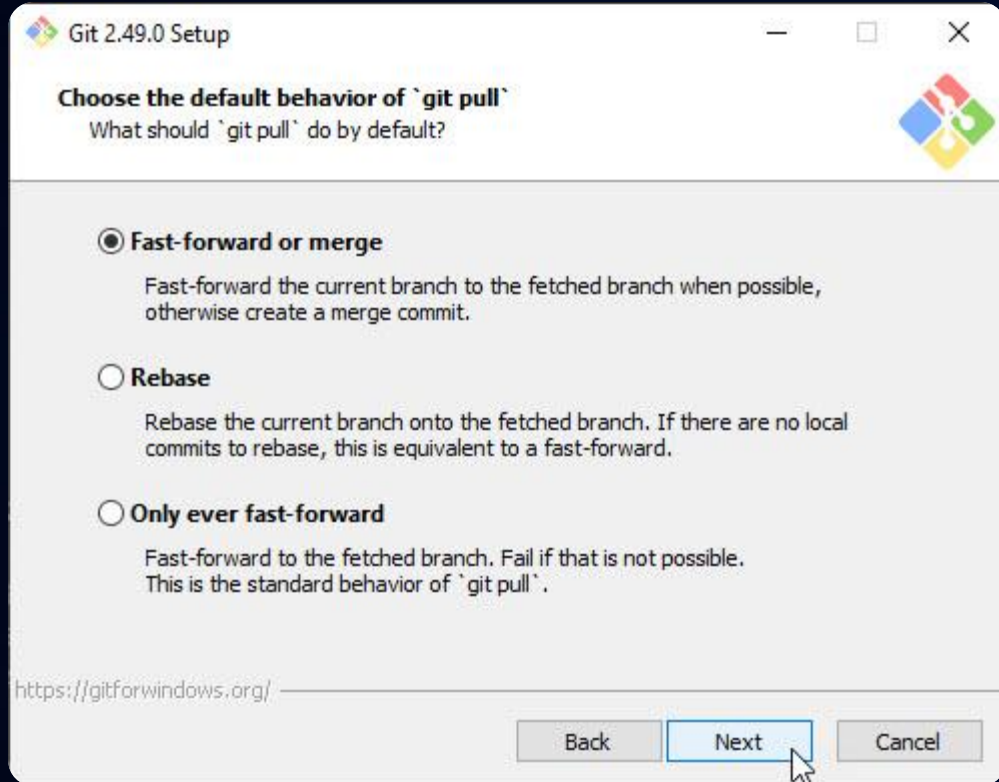


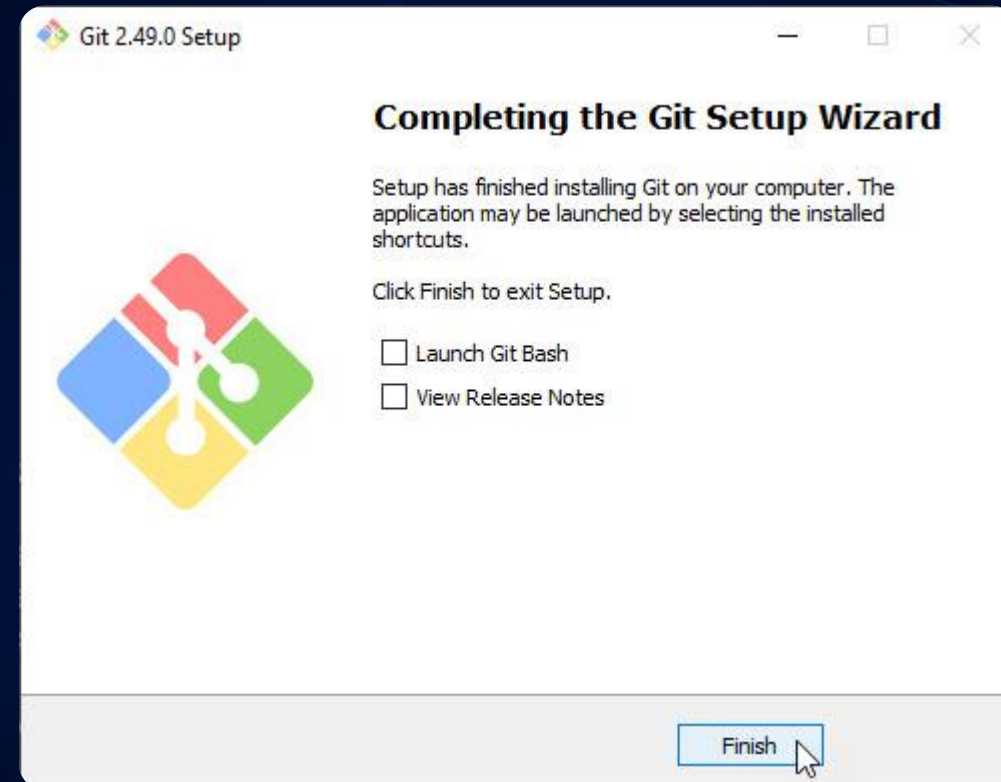
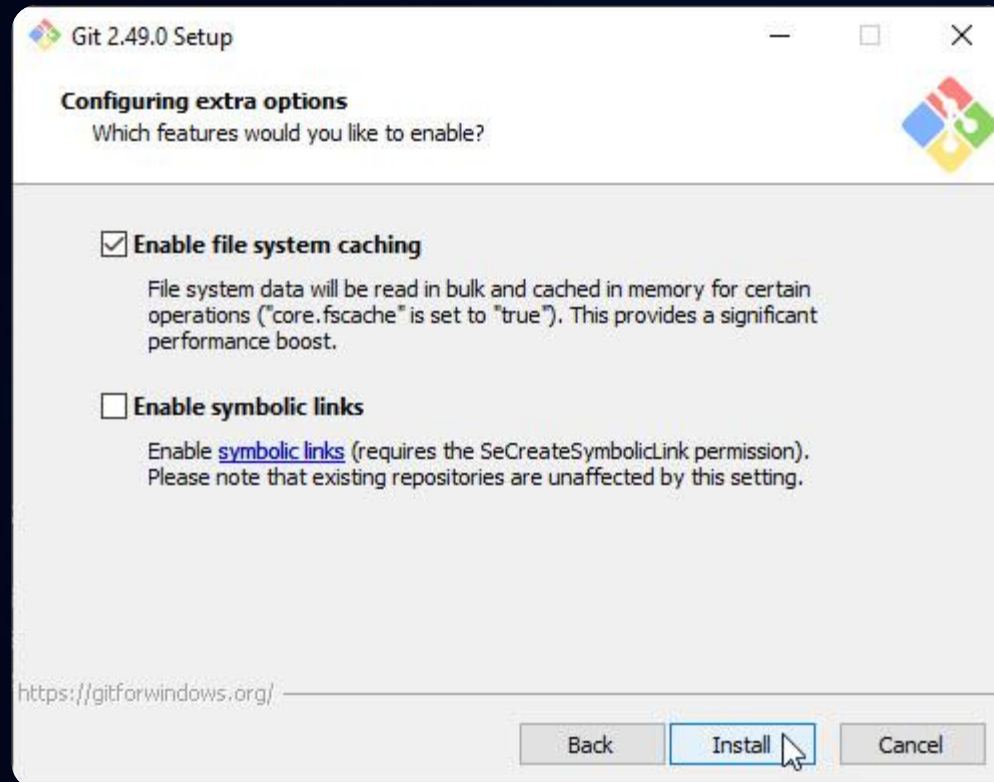










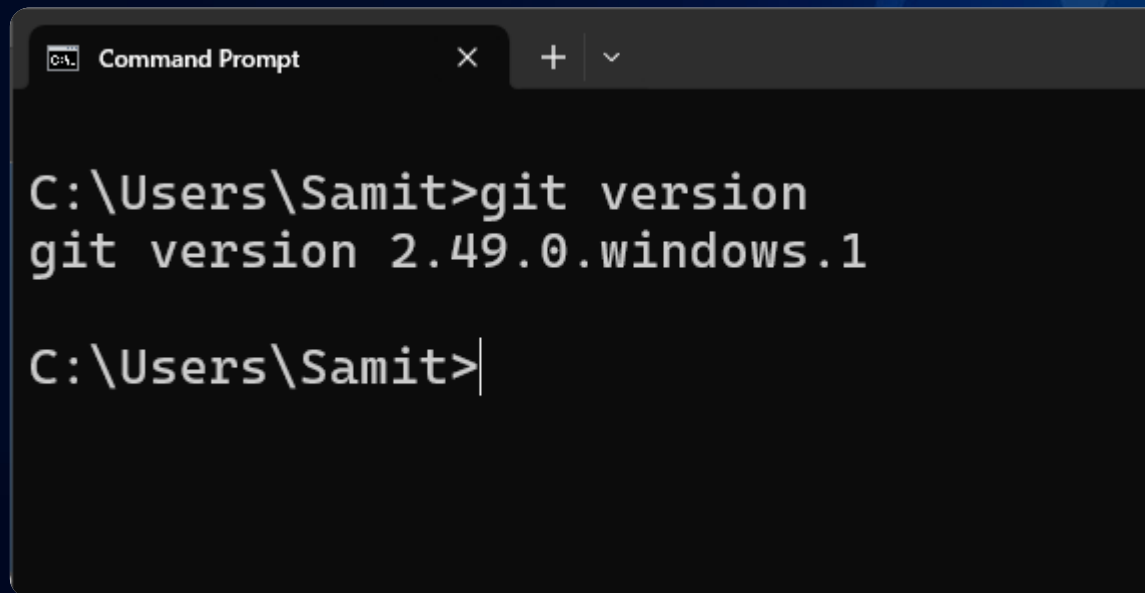




หลังติดตั้งสำเร็จทดสอบด้วยคำสั่ง

**git version**

หากพบเวอร์ชันดังภาพ ถือว่าติดตั้งเรียบร้อยแล้วพร้อมใช้งาน

A screenshot of a Windows Command Prompt window. The title bar shows 'C:\ Command Prompt' with standard window controls. The command prompt shows the user's location as 'C:\Users\Samit' and the command 'git version' has been entered and executed. The output is 'git version 2.49.0.windows.1'. The prompt is ready for the next command.

```
C:\Users\Samit>git version
git version 2.49.0.windows.1

C:\Users\Samit>
```

# การตรวจสอบความเรียบร้อยของเครื่องมือที่ติดตั้งทั้งหมด บน Windows / Mac OS / Linux

เปิด Command Prompt บน Windows หรือ Terminal บน Mac ขึ้นมาป้อนคำสั่งดังนี้

## Visual Studio Code

```
code --version
```

## Git

```
git version
```

## Node.JS

```
node -v  
npm -v  
npx -v
```

## Module 1.1

# React คืออะไร? (2026)

### Component-Based

แบ่ง UI เป็น Component เล็กๆ นำกลับมาใช้ซ้ำได้

### Declarative

บอกว่า UI ควรเป็นอย่างไร ไม่ใช่ทำทีละขั้น

### Virtual DOM

อัปเดตเฉพาะส่วนที่เปลี่ยน ทำให้เว็บเร็ว

### Open Source

สร้างโดย Meta, Community ใหญ่มาก

### npm downloads/week (approx. 2026)

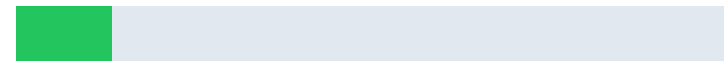
#### React

~30M/wk



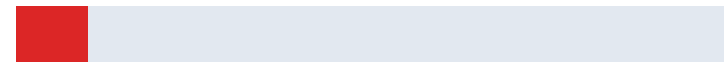
#### Vue

~4M/wk



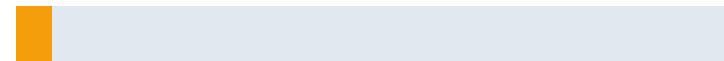
#### Angular

~3M/wk



#### Svelte

~1.5M/wk



### Used by:

Facebook

Instagram

Netflix

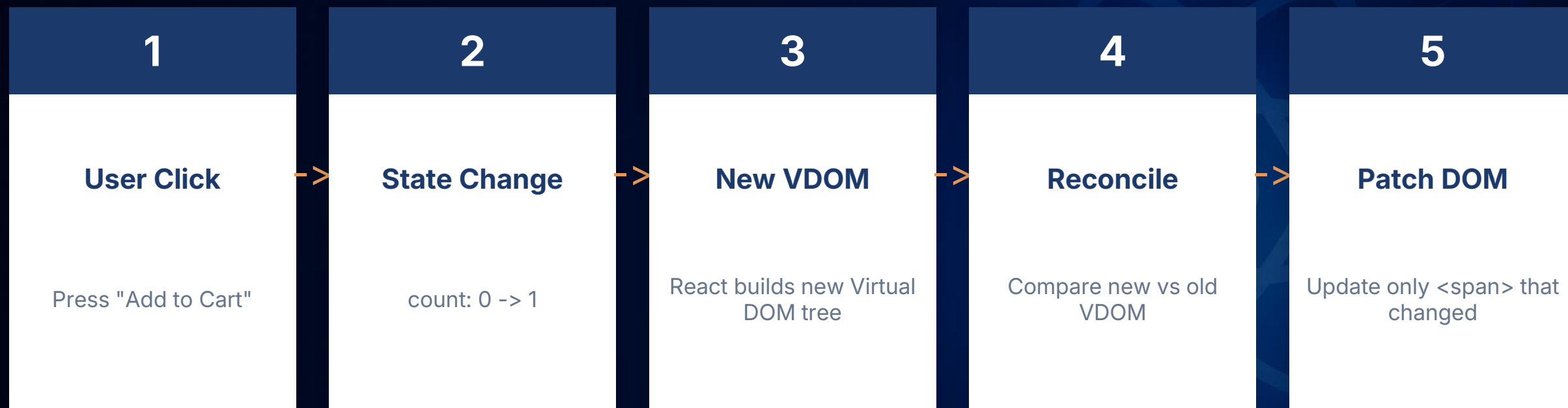
Airbnb

Discord

Uber



# Virtual DOM ทำงานอย่างไร?



## [KEY]

React does NOT re-render the whole page — it only updates the parts that actually changed.

## Module 1.1

# React Ecosystem 2026

[Build]

### Build Tool

Vite, Turbopack

[Frame]

### Meta-Framework

Next.js, Remix

[Route]

### Routing

React Router v7

[State]

### State Management

Zustand, Redux Toolkit

[Data]

### Data Fetching

TanStack Query v5 (React Query)

[Form]

### Forms + Validate

React Hook Form + Zod

[Style]

### Styling

Tailwind CSS v4, CSS Modules

[Test]

### Testing

Vitest, Testing Library

# ทำไมเลิกใช้ Create React App?

| Feature    | Create React App (Deprecated) | Vite — Standard 2026      |
|------------|-------------------------------|---------------------------|
| Status     | Deprecated 2023               | Active, Industry Standard |
| Dev Start  | 5–30 sec (slow)               | < 1 sec (super fast)      |
| HMR        | Slow (full reload)            | Instant (ES Modules)      |
| Bundler    | Webpack (legacy)              | Rollup + ESBuild          |
| TypeScript | Manual config required        | Built-in support          |

[TIP] In 2026, Vite is the industry standard. The React team recommends Vite over CRA.



## Module 1.3

# สร้างโปรเจกต์แรกด้วย Vite + TypeScript

1

```
npm create vite@latest my-react-app -- --template react-ts
```

สร้างโปรเจกต์พร้อม TypeScript template

2

```
cd my-react-app && npm install
```

เข้าโฟลเดอร์และติดตั้ง Dependencies

3

```
npm run dev
```

รัน Dev Server: <http://localhost:5173>

## Project Structure

```
my-react-app/  
├── src/  
│   ├── App.tsx # Main component  
│   └── main.tsx # Entry point  
└── vite.config.ts # Vite config
```

## HMR — Hot Module Replacement

Edit code -> Vite detects instantly  
-> Sends module via WebSocket  
-> Browser patches changed part  
-> State preserved! No full reload

## Module 1.4

# TypeScript ที่จำเป็นสำหรับ React

| JavaScript                 | TypeScript                      |
|----------------------------|---------------------------------|
| No Type                    | Explicit Types                  |
| Error at Runtime (Browser) | Error at Compile time (VS Code) |
| No Autocomplete            | Full Autocomplete               |
| Bugs hide easily           | Bugs appear while coding        |

## Basic Types

```
const name: string = 'Somchai'  
const age: number = 30  
const ok: boolean = true  
const nums: number[] = [90, 85]
```

## interface vs type

```
interface User {  
  id: number, name: string  
  age?: number // optional  
}  
type Status = 'active' | 'inactive'
```

## Module 1.4

# Generics & Union Types

## Generics — Works with any type

```
function getFirst<T>(arr: T[]): T { return arr[0] }
```

```
getFirst<number>([1,2,3]) // 1  
getFirst<string>(['a','b']) // 'a'
```

```
interface ApiResponse<T> {  
  data: T  
  status: number  
  message: string  
}  
// Usage:  
ApiResponse<Product[]>
```

## Union & Literal Types

```
type Direction = 'north' | 'south' | 'east' | 'west'
```

```
type ButtonVariant = 'primary' | 'secondary' | 'danger'
```

```
// Type Inference  
const count = 42 // number  
const name = 'React' // string  
const isOpen = false // boolean
```

[TIP] Use interface for Object shapes | Use type for Union Types and complex aliases

## Module 1.5

# JSX/TSX และ Function Component

JSX = Syntax Extension ที่เขียน HTML-like code ใน JavaScript | TSX = TypeScript + JSX

## JSX ที่เราเขียน

```
const el = <h1 className="title">  
  React!  
</h1>
```

## Compiled to JavaScript

```
React.createElement(  
  'h1', { className: 'title' },  
  'React!'  
)
```

## JSX Rules — Important Rules

[OK] Fragment: use `<> </>` instead of wrapping `<div>`

[OK] Close every tag: `<input />` `<img />` `<br />`

[OK] Use `className` instead of `class` attribute

[OK] Use `{}` for JS expressions: `{name}`, `{2+2}`

[OK] camelCase attributes: `onClick`, `tabIndex`, `onSubmit`

[OK] One root element per return — use Fragment `<> </>` to wrap

[NOTE] JSX is compiled away by Babel/TypeScript at build time — the browser only runs plain JavaScript.



## Module 1.5

# Function Component + Export / Import

## Welcome.tsx

```
interface WelcomeProps {  
  name: string  
  company: string  
}  
  
function Welcome({ name, company }: WelcomeProps) {  
  return (  
  
    <div><h2>Welcome {name}</h2></div>  
  )  
}  
export default Welcome
```

## Export / Import Patterns

```
// -- Default Export --  
export default function Button() {}  
  
import Button from './Button'  
  
// -- Named Export --  
export function Card() {}  
export function CardHeader() {}  
  
import { Card, CardHeader } from './Card'
```

## React Fragment — Avoid extra DOM nodes

```
// [OK] Fragment return ( <> <li>Item 1</li> <li>Item 2</li> </> )  
// [NO] Avoid return ( <div> <li>Item 1</li> <li>Item 2</li> </div> ) // extra DOM node
```

# สร้างหน้า "บัตรพนักงาน"

Build Employee Card — รวมทุกทักษะวันนี้

Skills used:

TypeScript

JSX/TSX

Props

Component

map()

1

## Create Project

```
npm create vite@latest employee-card -- --template react-ts
```

2

## Define Employee interface

```
interface Employee { id, firstName, lastName, position, department, isActive }
```

3

## Build EmployeeCard Component

```
function EmployeeCard({ employee }: { employee: Employee }) { ... }
```

4

## Render 3 cards in App.tsx

```
{employees.map(emp => <EmployeeCard key={emp.id} employee={emp} />)}
```

## Workshop

# Employee Interface & EmployeeCard Component

## src/types/employee.ts

```
export interface Employee {  
  id: number  
  firstName: string  
  lastName: string  
  position: string  
  department: string  
  email: string  
  startDate: string  
  isActive: boolean  
}
```

[NOTE] All fields in Employee interface are required. Use ? to make them optional.

## src/components/EmployeeCard.tsx

```
import type { Employee } from '../types'  
  
function EmployeeCard({ employee }: {  
  employee: Employee  
}) {  
  const status = employee.isActive  
    ? '[Active]' : '[Inactive]'  
  
  return (  
  
    <div>...{status}...</div>  
  )  
}  
export default EmployeeCard
```

## Usage in App.tsx

```
const employees: Employee[] = [ { id:1, firstName:'Somchai', ... }, ... ]  
{employees.map(emp => (<EmployeeCard key={emp.id} employee={emp} />))}  
// npm run dev -> http://localhost:5173
```

# สรุปวันที่ 1

## Basic to Intermediate React.js

[OK] React คืออะไร? Virtual DOM, Reconciliation, Ecosystem

[OK] Vite vs CRA — ทำไม Vite ถึงเป็นมาตรฐาน 2026

[OK] สร้างโปรเจกต์ด้วย Vite + React + TypeScript

[OK] TypeScript — Types, Interface, Generics, Union Types

[OK] JSX/TSX, Function Component, Export/Import, Fragment

[OK] Workshop: บัตรพนักงาน Arcelik Hitachi

Day 2  
Tomorrow

Component  
& Props

useState  
& Events

Conditional  
Render

Tailwind CSS v4



# Basic to Intermediate React.js



Thank you



อ.สามิตร โกยม  
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